```
for (int i = 0; i < s.length(); i++) {
    freq[s[i] - 'a']++;
    if freq[s[i] - 'a'] == 1) {
        q.push(s[i]);
    }

    while (!q.empty() && freq[q.front() - 'a'] > 1) {
        q.pop();
    }
    if (q.empty()) {
        res.push_back('#');
    } else {
        res.push_back(q.front());
    }
```

Last- Non-Repeating